Experiment 5

Student Name: Shubham Kumar
UID: 23BCS13887
Branch: CSE
Section/Group: KRG 3-B
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1. **Aim**: To design and analyze a scalable real-time messenger application (similar to WhatsApp / Facebook Messenger) that allows users to register, send one-to-one and group messages, share media, and receive messages instantly while ensuring high availability, reliability, and low latency using distributed system concepts.

2. Objective:

- To identify functional and non-functional requirements of a messaging system
- To design a high-level architecture (HLD) for a large-scale chat application
- To understand real-time communication using WebSocket
- To apply CAP theorem trade-offs in messaging systems
- To understand message queues, storage, caching, and delivery guarantees
- To study how availability and reliability are prioritized in chat systems

3. Tools:

- Draw.io – System Architecture & HLD Diagram
- MySQL / PostgreSQL – User & group metadata
- NoSQL DB (MongoDB / Cassandra) – Message storage
- Redis – Online presence & caching
- WebSocket Server – Real-time messaging
- Apache Kafka / Message Queue – Asynchronous delivery
- Object Storage (S3 equivalent) – Media files
- Load Balancer – Traffic distribution

4. System Requirements:

A. Functional Requirements:-

1) Client should be able to register and login securely
2) System should support one-to-one messaging
3) System should support group messaging
4) System should support text and media messages (image/video/file)
5) System should preserve chat history
6) Client should see message delivery and read receipts

7) Client should view chat list and conversation history
8) System should support creating and managing groups

B. Non-Functional Requirements:-

1. Scalability

- Target Users: 1 Billion users
- Avg Messages/User/Day: 100
- Total Messages: 100 Billion/day
- Storage Estimate: ~100 TB messages

System must handle millions of concurrent connections.

2. Availability vs Consistency (CAP Theorem)

Availability >>> Consistency (Eventual Consistency)

Reason:
- Messages must be delivered even during failures
- Slight delay in read receipts is acceptable
- Losing messages is unacceptable

Example:
- Message delayed → Acceptable
- Message lost → Critical failure

3. Latency

Target latency: 200–300 ms
Messages should appear almost instantly.

4. Reliability

- No message loss
- Guaranteed delivery
- Retry & acknowledgment mechanism

5. High Level Design (HLD):

The system follows a microservices-based architecture:

API Gateway

- Authentication, routing, rate limiting
- Entry point for all clients

User Service

- Registration & login
- Profile & contacts
- Stored in SQL DB

Messaging Service

- Handles sending/receiving messages
- Stores messages in NoSQL DB
- Maintains conversation threads

Real-Time WebSocket Server

- Persistent connections with clients
- Instant message delivery
- Presence & typing indicators

Group Service

- Group creation & membership
- Group metadata storage

Media Service

- Upload & download media files
- Stores media in object storage

Message Queue (Kafka)

- Asynchronous message delivery
- Decouples sender & receiver
- Ensures reliability

Redis Cache

- Online/offline status
- Recent chats
- Unread counts

Notification Service

- Push notifications for offline users

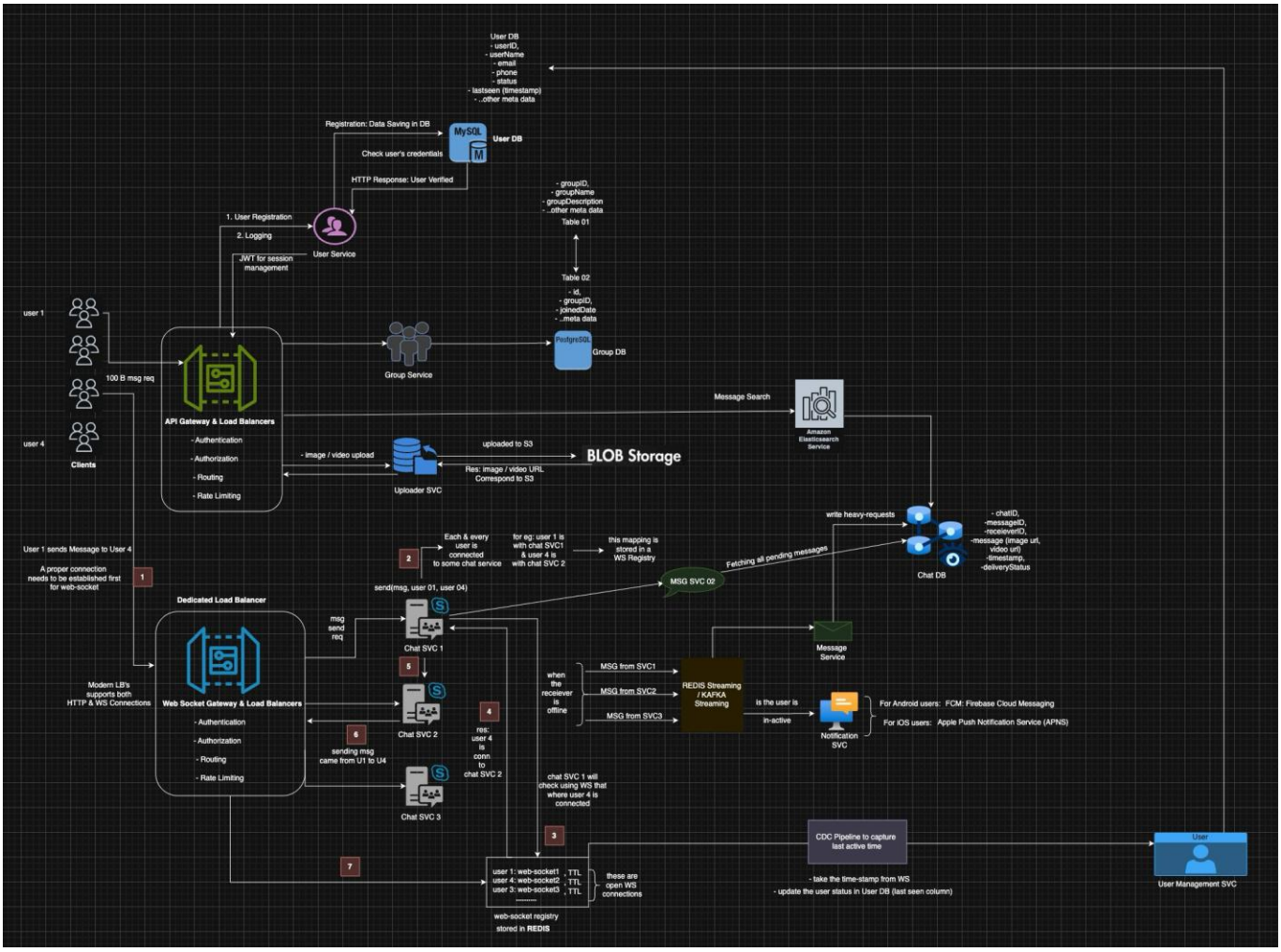
6. Messaging Workflow:

Step-1: User sends message
Step-2: Client → WebSocket Server

Step-3: Message stored in DB
Step-4: Message pushed to receiver if online
Step-5: If offline → Queue + Notification
Step-6: Receiver reads → Read receipt sent

7. Scalability Solution:

- Horizontal scaling of WebSocket servers
- Sharding messages by user ID
- Kafka queues for async delivery
- Redis caching for presence
- NoSQL DB for massive writes
- Load balancer distributes connections
- Stateless microservices



8. Learning Outcomes (What I Have Learnt):

- Understood how real-time messaging apps like WhatsApp are designed
- Learned difference between functional & non-functional requirements
- Gained understanding of WebSocket & real-time communication
- Learned CAP theorem trade-offs in chat systems
- Understood message queues & delivery guarantees
- Learned how to design highly available messaging systems